Weston Clark

Portland, OR | westoclark@gmail.com | github.com/westonclark | linkedin.com/in/westoclark | westonclark.dev

Skills

- **Languages**: TypeScript, JavaScript, C++, C, Golang
- **Technologies**: Node.js, PostgreSQL, Redis, Docker, Kubernetes, ArgoCD, GCP, AWS, RabbitMQ, Pub/Sub, BigQuery, React.js, Next.js, Jest, Vitest, CMake, JUCE

Work Experience

Software Engineer II May 2024 — Present

Madwire

- Designed the core APIs and databases for all user, account, and permission management across multiple products
- Developed the internal auth libraries and middleware that made up the user identity system alongside Keycloak
- Migrated legacy PHP microservices to Node.js and Golang, improving performance while reducing complexity and cost
- Shipped a custom role-based permission system for authorizing actions and resource access throughout the site
- Built a data warehouse ingest service with Pub/Sub and BigQuery to store data for reporting and LLM training
- Owned the custom notification services for enqueuing and sending SMS, email, and push notifications to users
- Created test suites and CI/CD pipelines with Jest and ArgoCD, optimizing for deployment times and compute cost

Audio Software Systems Engineer Jan 2019 — May 2024

Weston Clark Mixing LLC

- Contracted with AVL integrators and touring productions to design custom control software for live audio systems
- Wrote scripts and built embedded systems to parse timecode and trigger console snapshots via OSC messages over UDP
- Diagnosed and resolved clock-sync, latency, and packet-loss issues across networks before and during live shows
- Mixed front-of-house audio for arenas and amphitheater tours, performing for large audiences nightly

Technical Director Jan 2018 — Jan 2019

Westside AJC

- Led and staffed the production team of 12 audio/video operators for a church of 5,000 weekly attendees
- Trained non-technical volunteers to operate high-end, theatrical-level audio, video, and lighting equipment

Projects

Audio Mixing Engine github.com/westonclark/audio-engine

- A 16-channel C++ audio mixing engine that loads files, applies processing per-channel, and sums to a single output
- Separate threads for user input, audio processing, and background file loading, synced via atomics and lock-free queues
- Built directly on Apple's Core Audio API with no third-party libraries, using manual device and callback management

Fader Keys github.com/westonclark/fader-keys-source

- A desktop application for controlling fader movements in digital audio workstations via MIDI CC messages
- Deployed a custom online store and user dashboard for purchasing, downloading, and authorizing the application
- Application built with C++ and Objective-C++, website built with Next.js, Stripe, Clerk Auth, and Neon DB

HTTP Server github.com/westonclark/http-server-c

- An HTTP server written from scratch in C with no external libraries or frameworks
- Handles TCP socket connections, HTTP request parsing, and serves static HTML files from the file system

Multi Effect Audio Plugin github.com/westonclark/multi-effect-plugin

- An effect processor for applying DSP effects to audio recordings like phaser, chorus, overdrive, filters, and EQ
- Built using C++, the JUCE framework, and Xcode build tools

Watchdogs.io github.com/oslabs-beta/Watchdogs

- A web observability platform built for real-time monitoring of AWS Lambda performance metrics and logging
- Integrated with CloudWatch to retrieve performance data and display it using interactive dashboards and charts
- Designed an internal caching system with Redis to reduce third-party API costs and improve page load times